

End of Topic Test – Chapter 2: Personal Finance (Test 2)

Answer all questions in the spaces provided.

1. Budgeting and current account

A part-time worker keeps track of their money using a current account. Part of the statement for October is shown below.

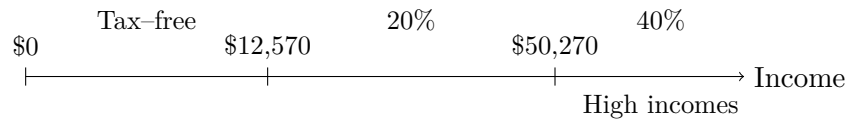
Date	Payment details	Paid out (£)	Paid in (£)	Balance (£)
30 Sep	Brought forward			45.60
01 Oct	DD Phone insurance	15.00		30.60
02 Oct	ATM Cash	40.00		<i>OD</i> 9.40
02 Oct	Unauthorised OD fee	6.00		<i>OD</i> 15.40
15 Oct	Employer salary		860.00	844.60
18 Oct	DD Car insurance	55.00		789.60
20 Oct	Card Online shopping	210.00		579.60
25 Oct	CHQ Rent	400.00		179.60
29 Oct	DD Electricity	48.75		130.85

- (a) (2 points) Explain what “OD” means on the statement, and why an “Unauthorised OD fee” was charged.
- (b) (4 points) Calculate the worker’s total income and total expenditure for October, using the statement.
- (c) (3 points) The worker wants to avoid overdraft charges in future. Suggest two realistic changes they could make to their budgeting, and explain how each would help.

2. Income tax bands

For a particular tax year, the income tax rules are:

- Personal allowance (tax-free): \$12 570
- Basic rate: 20% on taxable income from \$12 571 to \$50 270
- Higher rate: 40% on taxable income above \$50 270



- (a) (3 points) A worker has a gross annual income of \$45 200. Calculate their income tax for the year.
- (b) (4 points) Another worker has a gross annual income of \$76 000. Calculate their income tax for the year, clearly showing the tax paid in each band.
- (c) (2 points) For the worker in part (b), find their average (overall) tax rate as a percentage of their gross income, correct to 1 decimal place.

3. Payslip and deductions

A worker is paid monthly. Part of their payslip is shown below.

PAYSLIP – November	
Employee name	Jordan Lee
Employee number	03985
Gross pay	\$2 800.00
Income tax	\$350.60
National Insurance	\$195.84
Pension contribution	\$140.00
Net pay	\$2 113.56

- (a) (2 points) Explain the difference between gross pay and net pay.
- (b) (4 points) Calculate Jordan's net pay for November, showing your working.
- (c) (3 points) National Insurance is charged at 12% of monthly income between \$1 048 and \$4 189. Show whether the National Insurance deduction on the payslip is consistent with this rule.

4. Exchange rates

A family is converting British pounds (£) into euros (€) before a trip. Two options are available for converting £600:

- **Bureau A:** exchange rate $\text{£}1 = \text{€}1.14$, with a flat handling fee of £5 deducted from the pounds before conversion.
 - **Bank B:** exchange rate $\text{£}1 = \text{€}1.11$, with a 1.5% commission deducted from the pounds before conversion.
- (a) (2 points) Calculate the number of euros the family would receive from Bureau A, remembering to deduct the flat fee first.
- (b) (3 points) Calculate the number of euros the family would receive from Bank B, after deducting the 1.5% commission.
- (c) (3 points) Determine which option gives better value, and by how much (in euros).

5. Comparing loans using APR

A furniture shop offers a sofa for \$1 200 cash, or a “buy now, pay later” deal: a \$150 deposit now, followed by two payments of \$600 at the end of year 1 and at the end of year 2.

- (a) (3 points) State the amount that needs to be financed, and write down an equation (in terms of the annual interest rate i) that this amount must equal, once each future payment is discounted to its present value.
- (b) (3 points) Using trial and improvement, find the annual interest rate i implied by this deal, correct to the nearest 0.5%.
- (c) (3 points) The shop advertises this deal as “interest-free”. Comment on whether this description is accurate, using your answer to part (b).

6. Comparing savings using AER

A savings account pays a nominal annual interest rate of 3.6%, compounded monthly. A saver deposits \$1 500 into the account.

(a) (3 points) Find the monthly interest rate, giving your answer as a percentage and as a decimal.

(b) (3 points) Complete the table below to show the value of the savings at the end of each of the first three months. Give each value to the nearest penny.

	Starting (\$)	Interest (\$)	Final (\$)
Month 1	1500.00		
Month 2			
Month 3			

(c) (3 points) Calculate the Annual Equivalent Rate (AER) for this account, correct to 2 decimal places.

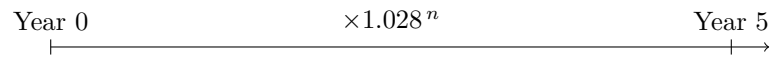
7. Depreciation

A delivery company buys a new van for \$28 000.

- (a) (3 points) The van depreciates by 15% of its value each year (reducing-balance depreciation). Calculate its value after 4 years, to the nearest dollar.
- (b) (3 points) An accountant instead models the van's value using straight-line depreciation of \$5 200 per year. Calculate its value after 4 years using this model.
- (c) (3 points) Compare the values found in parts (a) and (b), and explain which model is likely to be more realistic over a long period of time.

8. Inflation and real value

A classic car is currently worth \$45 000. Its value is projected to increase by 2.8% per year for the next 5 years.



- (a) (3 points) Write an expression for the projected value of the car after 5 years, and calculate this value to the nearest dollar.
- (b) (3 points) Suppose inflation is also 2.8% per year over the same 5 years. Explain briefly why the real (inflation-adjusted) value of the car has not changed.
- (c) (3 points) Over the same 5 years, general prices actually rose by a total factor of 1.152. Calculate the real growth factor of the car's value, and state the real percentage change (to 1 decimal place), including whether it is an increase or a decrease.

END OF TEST